

PALM BEACH INTL, FL, USA

WMO: 722030

Lat: 26.685N

Lon: 80.099W

Elev: 19

StdP: 14.69

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 12844

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 44.6	(c) 48.5	(d) 29.2	(e) 23.3	(f) 51.2	(g) 34.9	(h) 29.7	(i) 54.0	(j) 24.9	(k) 71.1	(l) 22.9	(m) 70.9	(n) 8.5	(o) 320	(p) 0.415

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 12.4	(c) 91.8	(d) 77.8	(e) 90.5	(f) 77.8	(g) 89.4	(h) 77.7	(i) 80.2	(j) 87.8	(k) 79.5	(l) 87.2	(m) 79.0	(n) 86.5	(o) 11.3	(p) 130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.1	146.2	83.7	77.4	142.6	83.5	76.9	140.2	83.2	43.5	87.8	42.9	87.0	42.4	86.6	84.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 21.6	(b) 19.5	(c) 18.1	(e) 38.0	(f) 94.4	(g) 3.4	(h) 1.5	(i) 35.5	(j) 95.5	(k) 33.5	(l) 96.4	(m) 31.6	(n) 97.3	(o) 29.1	(p) 98.4		
(a) 21.6	(b) 19.5	(c) 18.1	(e) 33.8	(f) 81.7	(g) 4.1	(h) 0.9	(i) 30.8	(j) 82.4	(k) 28.5	(l) 82.9	(m) 26.2	(n) 83.5	(o) 23.2	(p) 84.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	76.0	66.2	68.6	71.1	75.1	78.8	81.7	83.2	83.3	82.0	78.8	73.0	69.5
	DBStd	7.63	7.51	6.90	6.04	4.31	2.89	2.41	2.17	2.07	2.00	4.05	5.35	7.24
	HDD50	5	3	1	0	0	0	0	0	0	0	0	0	1
	HDD65	207	81	43	23	2	0	0	0	0	0	1	10	47
	CDD50	9488	505	523	653	752	894	950	1028	1033	961	894	690	605
	CDD65	4213	117	145	211	304	429	500	563	568	511	430	249	186
	CDH74	38109	438	659	1136	2042	3788	5185	6449	6536	5489	4001	1523	863
	CDH80	11958	21	69	188	391	1015	1837	2634	2658	1917	995	175	58
(14) Wind	WSAvg	9.1	9.4	9.5	10.5	10.5	9.6	7.9	7.4	7.4	7.9	9.7	9.6	9.3
(15) Precipitation	PrecAvg	60.2	3.1	2.7	3.5	3.3	5.4	8.1	5.8	6.9	8.6	6.1	4.0	2.4
	PrecMax	82.7	11.2	8.7	16.8	12.6	15.7	20.1	13.3	19.6	28.1	18.7	14.6	7.4
	PrecMin	37.8	0.1	0.2	0.4	0.0	0.4	1.7	1.2	1.7	1.8	0.6	0.2	0.1
	PrecStd	11.0	3.0	2.0	3.2	2.8	4.1	4.2	2.8	3.2	4.8	4.4	3.4	1.9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	82.4	85.0	87.7	89.1	91.2	93.3	93.4	93.3	91.2	89.2	85.9	83.6
		MCWB	71.7	71.2	72.9	73.6	74.7	78.4	77.9	78.5	77.7	76.7	75.0	74.1
	2%	DB	80.1	82.0	84.4	86.0	88.4	90.6	91.4	91.4	89.6	87.4	83.7	81.5
		MCWB	70.8	71.6	72.7	72.7	74.7	77.7	78.0	78.3	77.9	76.5	74.3	72.8
	5%	DB	78.3	80.2	81.9	84.0	86.7	89.1	90.1	90.2	88.5	86.2	82.1	80.0
		MCWB	69.7	70.8	71.5	72.1	74.4	77.4	78.0	78.3	77.6	75.9	72.8	71.7
	10%	DB	76.5	78.2	79.7	82.2	85.1	87.5	88.8	89.1	87.3	84.9	80.4	78.5
		MCWB	68.8	69.6	70.0	71.2	73.9	76.8	77.9	78.2	77.3	75.3	71.5	70.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	74.6	75.4	77.3	76.5	78.4	80.4	80.6	80.9	80.4	79.8	77.4	75.8
		MCDB	78.8	79.9	82.6	83.8	85.7	88.3	88.3	88.8	87.3	84.8	81.9	80.4
	2%	WB	73.0	73.5	74.9	75.2	77.3	79.4	79.9	80.2	79.7	78.5	76.1	74.6
		MCDB	77.5	78.3	80.7	82.0	84.1	87.1	87.9	88.1	86.6	84.0	80.9	79.2
	5%	WB	71.4	72.4	73.3	74.2	76.4	78.5	79.2	79.5	78.9	77.8	75.0	73.5
		MCDB	76.2	77.5	79.3	80.8	83.0	86.0	87.2	87.4	85.7	83.6	80.0	78.1
	10%	WB	70.2	71.2	71.9	73.1	75.5	77.9	78.5	78.9	78.3	77.0	73.7	72.4
		MCDB	75.3	76.5	78.0	79.7	82.0	85.3	86.2	86.6	84.9	82.8	78.7	77.1
(35) Mean Daily Temperature Range	MDBR		15.8	15.5	15.0	13.6	12.9	12.3	12.4	12.4	11.5	11.4	13.3	13.9
	5% DB	MCDBR	14.1	14.9	15.7	15.1	15.0	13.9	13.3	13.4	12.5	11.6	12.3	12.2
		MCWBR	7.8	6.9	7.2	6.3	5.9	5.6	5.3	5.1	4.9	4.9	5.9	6.4
	5% WB	MCDBR	12.9	13.3	14.6	13.4	13.3	13.0	12.6	12.4	11.7	10.7	10.8	11.1
		MCWBR	8.0	7.1	7.8	6.2	5.9	6.0	5.5	5.4	5.2	5.0	6.0	6.8
(40) Clear-Sky Solar Irradiance	taub		0.359	0.367	0.384	0.400	0.431	0.460	0.473	0.463	0.431	0.412	0.379	0.367
	taud		2.522	2.516	2.453	2.391	2.344	2.327	2.280	2.334	2.427	2.441	2.485	2.518
	Ebn at Noon		283	289	289	285	275	265	261	264	271	270	273	274
	Edh at Noon		29	32	36	39	41	41	43	40	36	34	30	28
(44) All-Sky Solar Radiation	RadAvg		1123	1377	1681	1937	1975	1829	1839	1774	1556	1425	1197	1033
	RadStd		77	87	97	88	166	169	104	100	83	102	62	74

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.72	+1.55	N/A	N/A	N/A	N/A	N/A	-49	+241	+193
(47) Regional (27 neighbors)		+0.70	+1.53	N/A	N/A	N/A	N/A	N/A	-38	+248	+202

Nomenclature: See separate page